Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Driver 3 and matrix

File: matrix3.kicad_sch

Driver 4 and matrix

File: matrix4.kicad_sch

Square Rounded-Edge Bezel WLCSP-36 (bga13-169): 169 MHPA0808RGBDT 0808 LEDs (13x13 @ 1.50 mm in 38x38 mm R3 square bezel), 4x IS31FL3745 WLCSP-36 behind array, USB-C + JST 1.25, 8L ENIG.
38 x 38 mm (R=3.0 mm) 8-layer ENIG PCB: 4x IS31FL3745 WLCSP-36 BGA drivers sit on B.Cu directly behind the centered LED array;
RP2350A, USB-C (TYPE-C-31-M-22), Micro-JST 1.25 (A1251WR-S-2P), flash, and charger sit around the 4-sided bezel.

Drawn by scripts/square_bezel/sch_gen.py

169 MHPA0808RGBDT 0808 LEDs (13x13 @ 1.50 mm in 38x38 mm R3 square bezel), 4x IS31FL3745 W

CERN-OHL-P-2.0

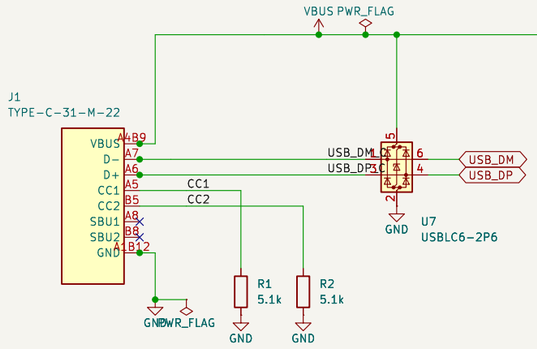
Sheet: /

File: dense-square.kicad_sch

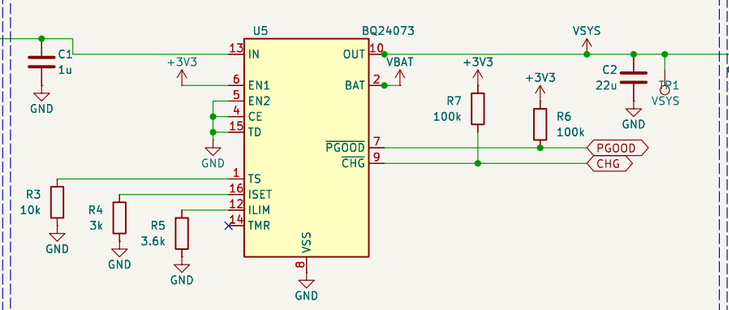
Title: dense-square (bga13-169): top

Size: A4	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 1/7

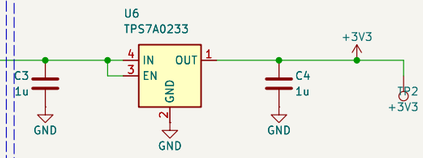
USB-C 16P SMT & ESD



CHARGER AND POWER PATH

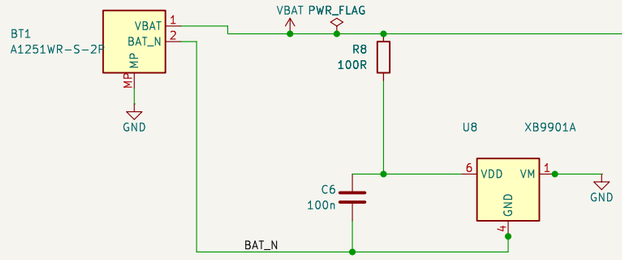


3V3

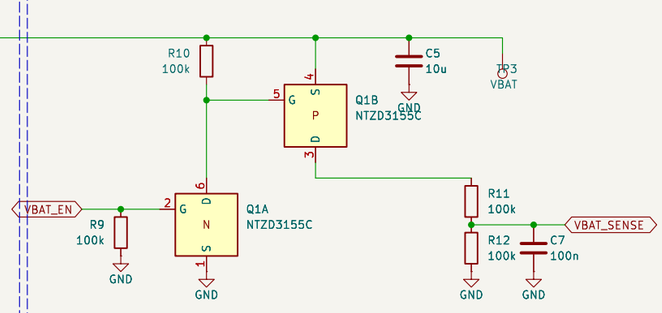


CELL AND PROTECTION

Micro-JST 1.25 / MX1.25 2P SMT (1=VBAT+, 2=BAT_N-).



VBAT SENSE



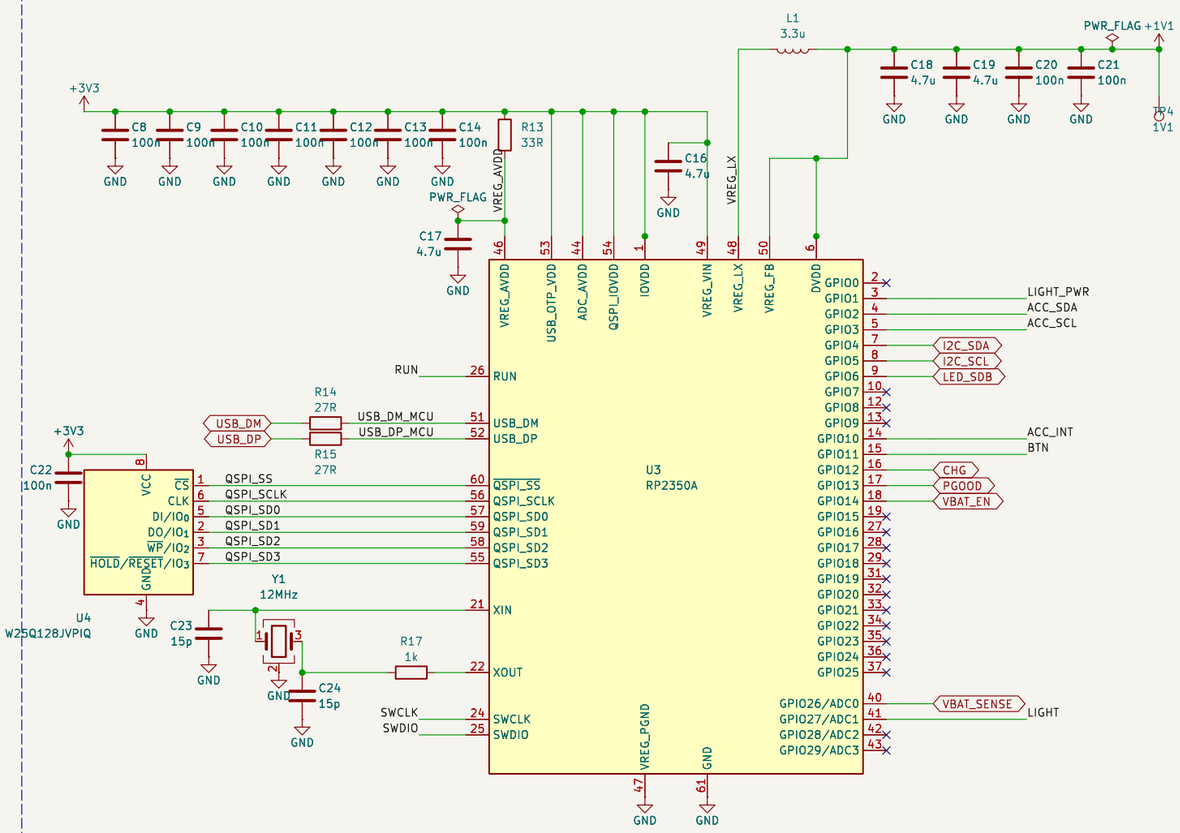
Drawn by scripts/square_bezel/sch_gen.py
USB-C 16P SMT, ESD, BQ24073 charger, Micro-JST 1.25, XB9901A, LDO, VBAT sense
CERN-OHL-P-2.0

Sheet: /Power/
File: power.kicad_sch

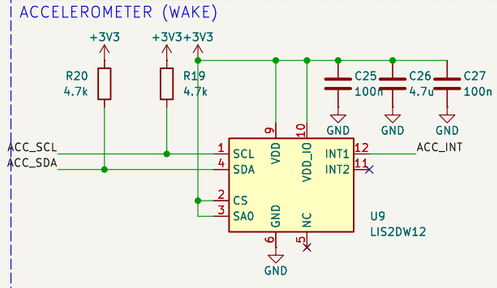
Title: dense-square (bga13-169): power

Size: A3	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 2/7

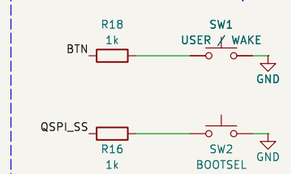
RP2350A, FLASH, CRYSTAL



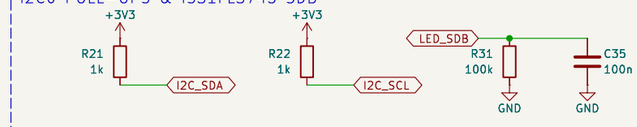
ACCELEROMETER (WAKE)



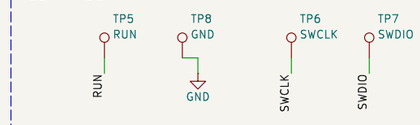
PEG-LESS SIDE BUTTONS (EVQP7J01P)



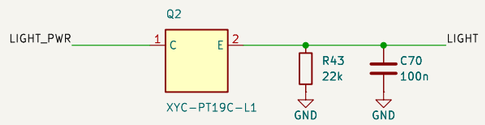
I2C0 PULL-UPS & IS31FL3745 SDB



TEST PADS



AMBIENT LIGHT SENSOR (NORTH BEZEL)



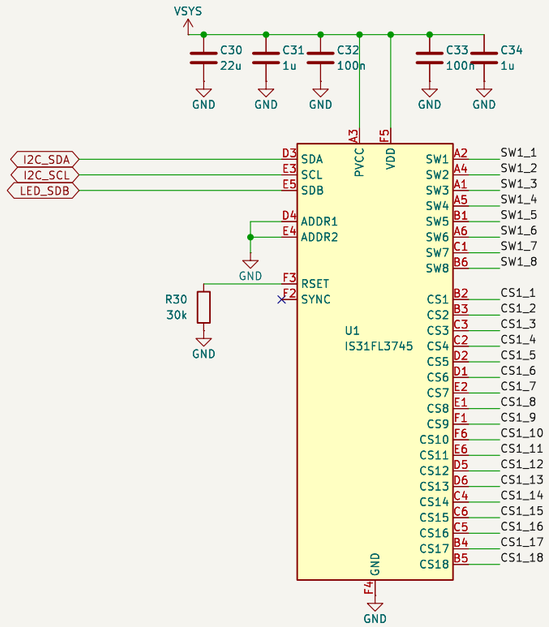
Drawn by scripts/square_bezel/sch_gen.py
RP2350A, flash, crystal, accelerometer, side buttons, light sensor, I2C0 & LED_SDB
CERN-OHL-P-2.0

Sheet: /MCU/
File: mcu.kicad_sch

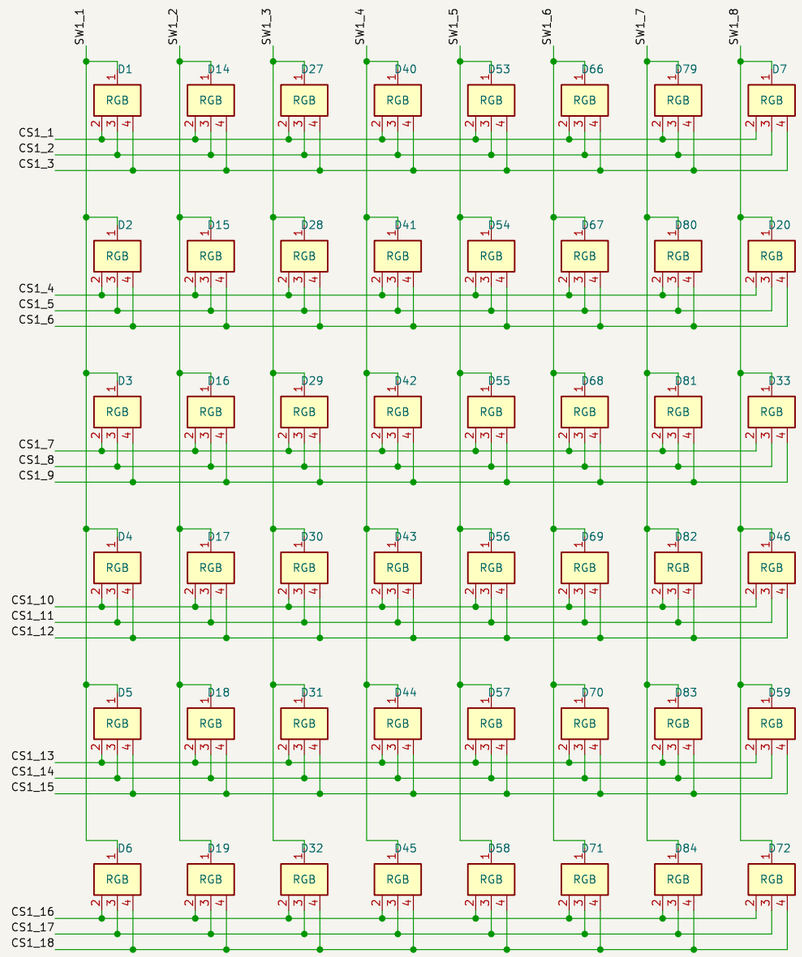
Title: dense-square (bga13-169): mcu

Size: A3	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 3/7

DRIVER 1 (U1, 0x20)



48 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



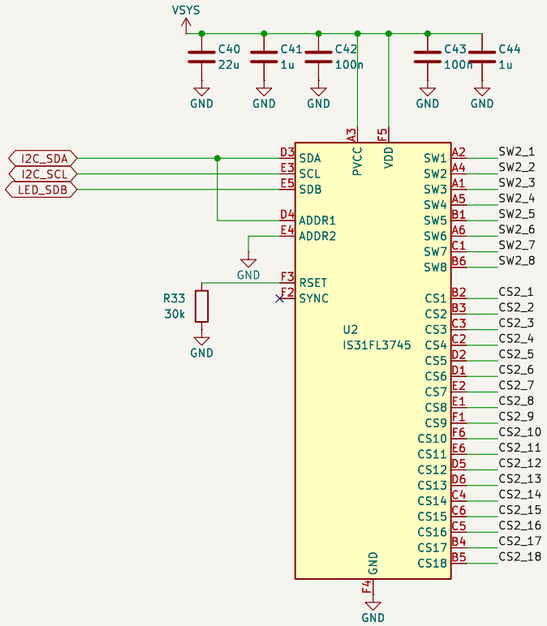
Drawn by scripts/square_bezel/sch_gen.py
U1 IS31FL3745-CLS4-TR WLCSP-36 (0x20), 48 MHPA0808RGBDT 0808 LEDs
CERN-OHL-P-2.0

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

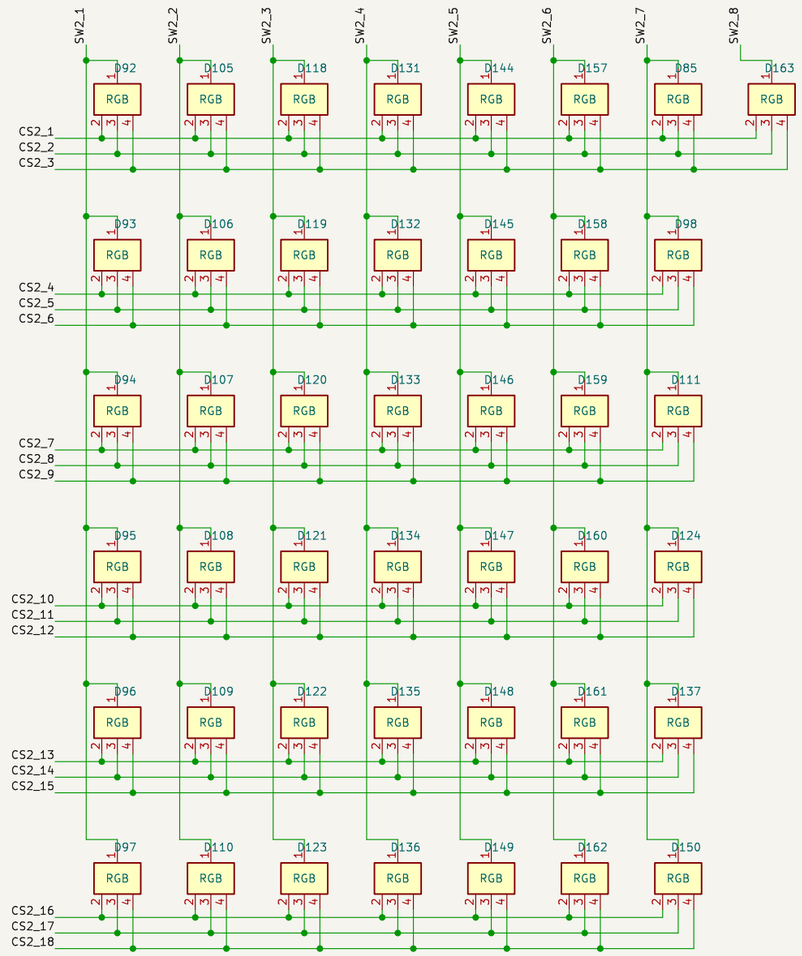
Title: dense-square (bga13-169): driver 1 and matrix

Size: A3	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 4/7

DRIVER 2 (U2, 0x22)



43 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



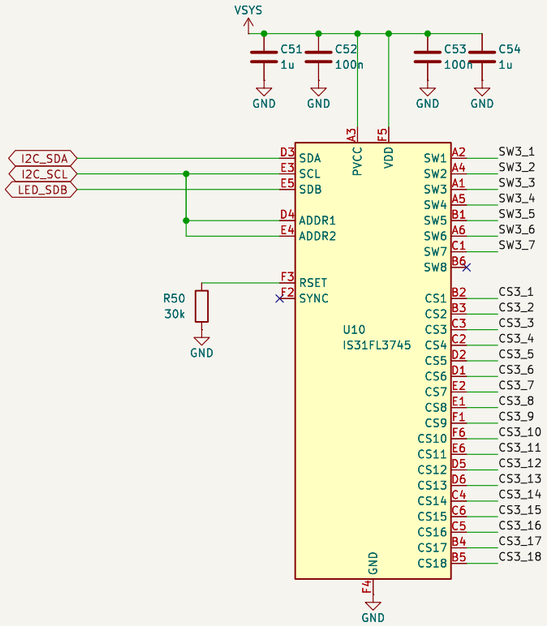
Drawn by scripts/square_bezel/sch_gen.py
U2 IS31FL3745-CLS4-TR WLCSP-36 (0x22), 43 MHPA0808RGBDT 0808 LEDs
CERN-OHL-P-2.0

Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

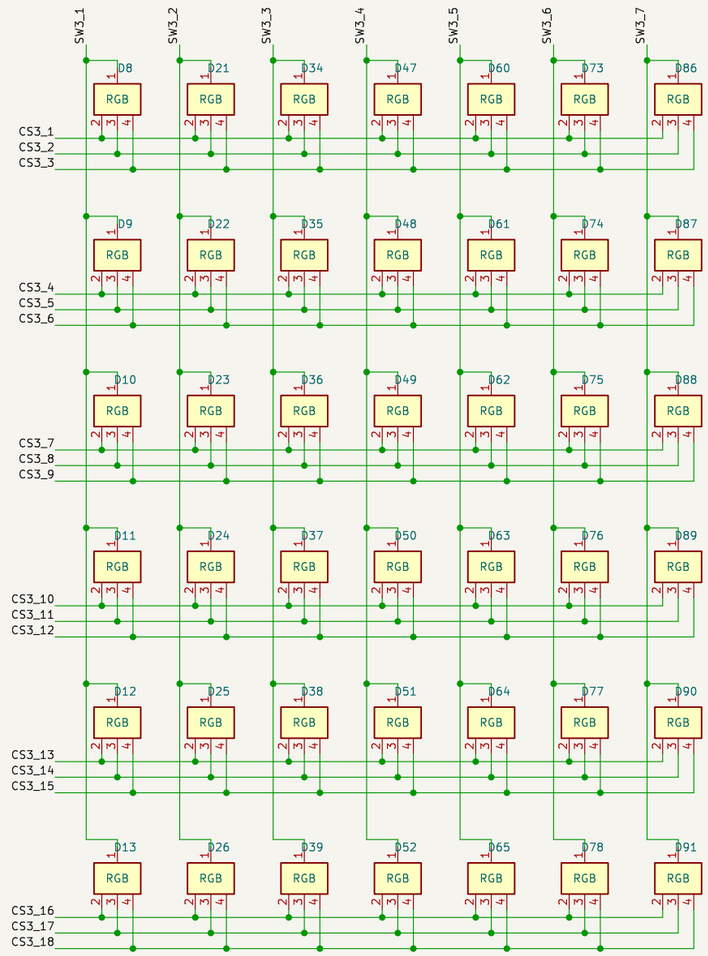
Title: dense-square (bga13-169): driver 2 and matrix

Size: A3	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 5/7

DRIVER 3 (U10, 0x25)



42 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



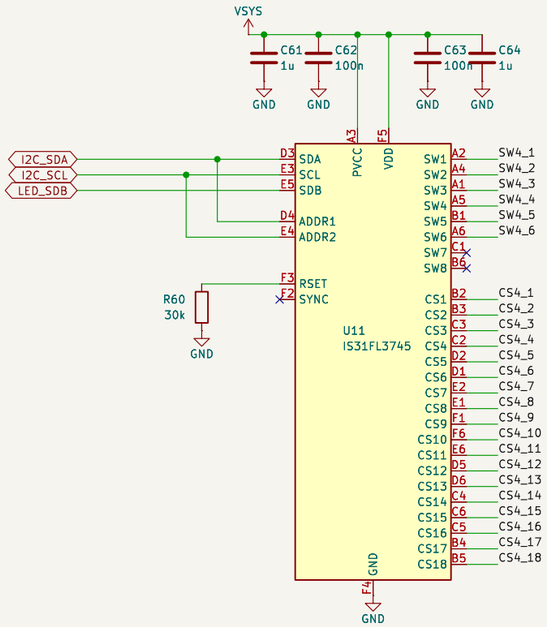
Drawn by scripts/square_bezel/sch_gen.py
U10 IS31FL3745-CLS4-TR WLCSP-36 (0x25), 42 MHPA0808RGBDT 0808 LEDs
CERN-OHL-P-2.0

Sheet: /Driver 3 and matrix/
File: matrix3.kicad_sch

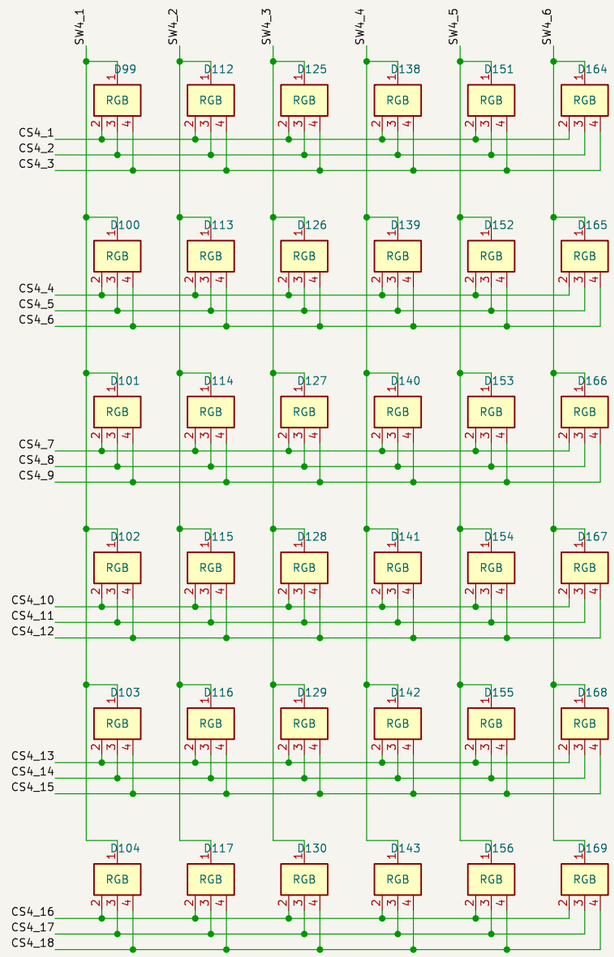
Title: dense-square (bga13-169): driver 3 and matrix

Size: A3	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 6/7

DRIVER 4 (U11, 0x26)



36 MHPA0808RGBDT 0808 LEDS: A=Pin1, R=Pin2, G=Pin3, B=Pin4



Drawn by scripts/square_bezel/sch_gen.py
U11 IS31FL3745-CLS4-TR WLCSP-36 (0x26), 36 MHPA0808RGBDT 0808 LEDs
CERN-OHL-P-2.0

Sheet: /Driver 4 and matrix/
File: matrix4.kicad_sch

Title: dense-square (bga13-169): driver 4 and matrix

Size: A3	Date: 2026-11-06	**Rev: SB13**
KiCad E.D.A. 9.0.8		Id: 7/7